Acoustic drive of a sonoluminescence flask

Electrical → mechanical → acoustic in one circuit, in H_2O and D_2O — CLAUDE.md §3.1, §3h

kronspice simulation report · 2026-11-07 · AC sweep, 2001 points over ±3 % of f_0

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Summary

One netlist joins three energy domains of CLAUDE.md §3.1: a 50 Ω generator, an optional series inductor that tunes out the transducer's capacitance, a piezoelectric ring, its coupling into the breathing mode of a 3 cm liquid sphere, and that mode itself, in light water (H_2O) and heavy water (D_2O). The question is how much drive and power are needed for a 1.3 atm pressure amplitude at the centre of the flask, the range where single-bubble sonoluminescence is observed, and where every watt goes.

- **Power into the liquid is fixed by the mode:** 6.96 mW in H_2O and 6.72 mW in D_2O at 1.3 atm, equal to $P^2/(2 \cdot R_{par})$, with or without matching.
- **Matching raises the pressure per volt about 21 times:** from 3.22 to 69.4 kPa/V in H_2O, so the generator EMF falls from 40.9 V to 1.90 V.
- **Matching moves the loss, it does not remove it:** without the inductor the transducer's loss takes about a third of the input (3.50 of 10.9 mW); with it, most of the input (19.0 of 26.2 mW) is burned in the generator's own 50 Ω.
- **D_2O resonates 6.5 % lower** than H_2O (23.40 against 25.03 kHz; sound speed 1404 against 1502 m/s) and takes slightly less power.
- **The ledger closes in every run:** source power in equals heat out, residual 0 of the scale; the tests require below 10^{-9}.

What is being tested

This is a standard-physics construction, not a register hypothesis. It is the first circuit that joins the electrical, mechanical and acoustic channels through Kron's network analogies (§3a.1 items 3 and 7): each non-electrical channel has an across and a through variable whose product is power, and different domains are joined by connection tensors (Kron 1939, Ch. V p. 133). The liquids come from the fluid/ package (§3h). The bubble itself is out of scope here; it enters as a cited equivalent circuit in examples/bubble_collapse, on this chain's pressure node p.

Sources: Brenner, Hilgenfeldt & Lohse, Rev. Mod. Phys. 74, 425 (2002); Kinsler et al., Fundamentals of Acoustics, 4th ed. (2000), §7.5 and §9.3; Firestone, J. Acoust. Soc. Am. 4, 249 (1933); Wagner & Pruß, J. Phys. Chem. Ref. Data 31, 387 (2002); Herrig et al., J. Phys. Chem. Ref. Data 47, 043102 (2018); Kron, Tensor Analysis of Networks (1939), Ch. V p. 133.

The model

Every node belongs to one domain, and ground is the common reference of all of them (zero velocity, ambient pressure). The mechanical domain uses the mobility analogy (velocity across, force through), so a mass is a capacitor to ground and Kirchhoff's current law is the force balance at a point. The acoustic domain is pressure across, volume velocity through.

stage	elements	domain
generator	VGEN 1 V AC, RGEN 50 Ω	electrical
matching	LMATCH = $1/(\omega_0^2 \cdot C_0)$ with 0.5 Ω, or RWIRE 0.5 Ω alone	electrical
piezo ring, electrical side	C_0 4 nF clamped capacitance	electrical
transducer	gyrator YPZT, force factor φ = 0.6 N/V	electrical ↔ mechanical
piezo ring, mechanical side	CMASS 4 g moving mass; LSPRING 1/k with k = $\omega_0^2 \cdot m$ (resonance placed at the mode); RLOSS 1/b, b = 3 N·s/m	mechanical

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stage	elements	domain
coupling	gyrator YCOUPLE, effective area S_{eff} = 2 cm²	mechanical ↔ acoustic
flask breathing mode	CMODE C_a, LMODE M_a, RMODE R_{par} from fluid.SphericalResonator	acoustic

A transducer is an ideal gyrator, $i_1 = g \cdot V_2$, $i_2 = -g \cdot V_1$, which passes power from one port to the other without loss: a piezo element of force factor φ has g = φ, a piston of area S has g = S. In Kron's primitive network it is two axes coupled by an antisymmetric mutual admittance.

The breathing mode of a liquid sphere of radius R with a pressure-release wall sits at $f_0 = c/(2R)$. It is a parallel acoustic resonator whose across variable is the centre pressure: $C_a = 2R^3/(\pi\rho c^2)$ from the mode's potential energy, $M_a = 1/(\omega_0^2 \cdot C_a)$, $R_{par} = Q/(\omega_0 \cdot C_a)$, with an assumed Q of 1500. The program sweeps 2001 points over ±3 % of f_0 with analysis.AC, finds the pressure peak, scales the drive to 1.3 atm at the centre and reads the ledger.

Results

All four runs at a 1.3 atm centre amplitude (the sonoluminescence range is about 1.2–1.5 atm at 20–40 kHz, Brenner, Hilgenfeldt & Lohse 2002). f_0 is the mode frequency; the pressure peak of the chain is in brackets.

run	f_0 (peak)	pressure per volt	generator EMF	source power	into the liquid	transducer loss	generator's own 50 Ω
H_2O, no matching	25.03 kHz (24.27)	3.22 kPa/V	40.9 V	10.9 mW	6.96 mW	3.50 mW	0.42 mW
H_2O, matched	25.03 kHz (25.02)	69.4 kPa/V	1.90 V	26.2 mW	6.96 mW	0.42 µW	19.0 mW
D_2O, no matching	23.40 kHz (22.69)	3.20 kPa/V	41.2 V	10.5 mW	6.72 mW	3.26 mW	0.51 mW
D_2O, matched	23.40 kHz (23.40)	72.2 kPa/V	1.82 V	23.5 mW	6.72 mW	0.39 µW	16.6 mW

- Without the inductor the chain's peak sits below f_0 (24.27 kHz in H_2O) and the transducer loss takes a third of the input. With it the chain is a low impedance at resonance, most of the input is dissipated in the generator's 50 Ω, and the peak returns to f_0.

Ledger and validation

- Ledger: source power in equals heat out in every run, residual 0 of the scale.
- main_test.go checks, for both liquids: liquid heat equals $P^2/(2 \cdot R_{par})$ to 10^{-9}; ledger residual below 10^{-9}; matching puts the peak back at f_0 within 0.1 % and raises the pressure per volt by more than ten times; D_2O's f_0 is below H_2O's.
- Kron twin (export sheets): the matched chains are torn automatically at LMATCH into two pieces; torn and MNA node voltages agree to 4.6×10^{-12} (H_2O) and 2.6×10^{-12} (D_2O) of the largest node voltage, and the power $V \cdot I^*$ is the same torn and untorn (Roth 1959 §5). The unmatched H_2O chain is torn at RGEN and agrees to 4.4×10^{-15}.

Limits of the model

- The flask Q (1500), the coupling area S_{eff} and the transducer values (C_0, φ, m, b) are declared assumptions in main.go, not measurements. A hydrophone calibration of the real flask sets them; after that the chain is a design tool, and an evolvable one (matching network, transducer and drive frequency for the least power at a target pressure).
- The bubble dynamics (Rayleigh–Plesset or Keller–Miksis, collapse, light) stay out of the engine (CLAUDE.md §1, non-goals); they enter as a cited equivalent circuit in examples/bubble_collapse.
- These acoustic modes, about 25 kHz, sit ten orders of magnitude below molecular vibration frequencies (doc/future-work.md).

Reproducing the results

```
go run ./examples/acoustic_drive
go test ./examples/acoustic_drive ./fluid
python examples/acoustic_drive/viz/make_report.py examples/acoustic_drive/acoustic_drive_report.pdf
```


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